HWO Software Development & ICT Consultancy
ENTERPRISE SOLUTIONS · GOVERNMENT · WORLD BANK PROJECTS

SYSTEM BLUEPRINT · CONFIDENTIAL

PIIMS

A unified platform for World Bank–financed project inception, results tracking, financial oversight, and impact evaluation — delivered for the Ministry of Water and Energy, Federal Democratic Republic of Ethiopia.

DOCUMENT
System Blueprint v2.0

DATE
October 2026

DEVELOPED BY
HWO Software Development & ICT Consultancy

CLIENT
Ministry of Water and Energy (MoWE)

FUNDER
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CLASSIFICATION
Restricted

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Version	Date	Author	Change
1.0	2025	HWO Solutions Team	Initial baseline (17 modules, PostgreSQL)
2.0	October 2026	HWO Solutions Team	Impact-enabled revision — 23 modules, MySQL 8.4, full RBAC, 10 integrations

About HWO Software Development & ICT Consultancy

HWO Software Development & ICT Consultancy is a software engineering and ICT consulting firm specializing in enterprise applications for government institutions, development partners, and international financial institutions. This blueprint reflects our standard delivery methodology: production-grade architecture, security-first design, comprehensive documentation, and full handover to client IT teams.

Delivery Approach

Phased delivery with iterative milestones, full source code, documentation, and knowledge transfer to client teams.

Technical Discipline

Production-grade stack, security-first design, comprehensive test coverage, and observable systems by default.

Handover

Complete documentation, runbooks, training materials, and a 90-day post-handover support window.

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> **Reading guide.** This blueprint is an architectural overview intended for stakeholder review. For implementation details, refer to the SAD/SDD, the API Specification, and the Data Dictionary.

01 Executive Summary

PIIMS is the World Bank–financed project inception information system for the Ministry of Water and Energy. It replaces fragmented spreadsheets and manual reporting with a single governed platform spanning the entire project lifecycle — from idea to impact evaluation.

What PIIMS Delivers

- **Single source of truth** for project inception data across MoWE, MoF, DBE, and the World Bank
- **23 integrated modules** covering projects, results, finance, procurement, beneficiaries, grievances, and impact
- **43 user roles** across 14 stakeholder groups, with fine-grained RBAC
- **Full impact measurement** from indicator design → field collection → QA → warehouse → research access → analysis
- **10 external integrations**: FMIS, DBE, WB, SMS, USSD, Mobile Money, GIS, IdP, ODK, Safe Room
- **Bilingual** English + Amharic, offline-capable, and audit-ready

At a Glance

23 MODULES	**62** DATABASE ENTITIES	**43** USER ROLES	**10** EXTERNAL INTEGRATIONS
14 STAKEHOLDER GROUPS	**2** LANGUAGES (EN, AM)	**99.5%** UPTIME TARGET	**20 yr** IMPACT DATA RETENTION

Why It Matters

For the Ministry

Real-time portfolio oversight, standardized reporting, and a defensible record for every project decision.

For the World Bank

Measurable results, auditable fiduciary data, and a research-grade impact evaluation pipeline.

For Beneficiaries

Faster service delivery, SMS-based registration, and a functioning grievance redress mechanism.

For the Future

Longitudinal data that enables credible impact evaluation for years after project close.

02 System Context & Stakeholders

External Systems & Data Flows

Financial Systems

MoF IFMIS · budget & expenditure DBE Loan Mgmt · on-lending World Bank · ISR/PPMR

Communication

SMS Gateway · beneficiary SMS USSD Gateway · feature phones Mobile Money · payments

Field & Spatial

ODK Central · survey forms GIS Platform · basemaps & boundaries Government IdP · SSO

Research

Safe Room · anonymized datasets External researchers · DUA-controlled

Stakeholder Matrix

Stakeholder	Role in System	Users	Key Responsibilities
Ministry of Water & Energy	System owner, PIU, M&E, Impact Evaluation	9 roles	Data entry, results tracking, impact measurement
Ministry of Finance	Budget oversight, financial control	3 roles	Budget validation, expenditure monitoring, audit
Development Bank of Ethiopia	On-lending intermediary	4 roles	Loan management, disbursement, portfolio tracking
World Bank	Financier, supervision, impact evaluation	6 roles	Portfolio review, no-objection, ISR data
Solar Suppliers	Service provider	3 roles	Delivery submission, invoices, warranty
Local Government	Field verification	4 roles	Beneficiary registration, verification, geo-tagging
Data Collectors	Field data collection	2 roles	CAPI interviews, offline sync
Beneficiaries	End recipients	2 roles	SMS self-registration, service confirmation, grievance
Researchers	Impact evaluation	1 role	Dataset access under DUA
Auditors / Evaluators	Independent oversight	2 roles	Read-only audit trail access

Stakeholder	Role in System	Users	Key Responsibilities
Steering Committee	Governance	1 role	Portfolio review via dashboards
System / Support	Infrastructure	4 roles	DBA, DevOps, Security, Help Desk

> **43 roles** across **14 stakeholder groups** — each with fine-grained permissions, organizational scoping, and geographic data boundaries.

03 Architectural Overview

*PIIMS follows a **modular monolith** with a service-oriented integration layer, deployed on a layered architecture. This balances simplicity of operations with clean module boundaries and a future migration path to microservices.*

Frontend Architecture

Layer	Technology	Purpose
Framework	React 18 + TypeScript 5 (strict)	Type-safe, component-based UI
Bundler	Vite 5 + PWA plugin	Fast builds, offline support
State	TanStack Query + Zustand	Server cache + lightweight client state
Styling	Tailwind CSS 3 + design tokens	Consistent, accessible design system
i18n	react-i18next (EN + AM)	Full bilingual support
Forms	React Hook Form + Zod	Type-safe validation
Maps	Leaflet + react-leaflet	Interactive GIS
Testing	Vitest + Testing Library + Playwright	Unit + E2E coverage

Backend Architecture

Layer	Technology	Purpose
Runtime	Node.js 20 LTS	Async, high-concurrency workloads
Framework	Express 4 + TypeScript 5 (strict)	Proven, mature HTTP layer
Database	MySQL 8.4 + Prisma 5	Relational data integrity, migrations
Cache	Redis 7	Sessions, cache, idempotency
Queue	BullMQ	Reports, notifications, sync jobs
Storage	S3-compatible (MinIO)	Documents, media, datasets
Auth	JWT + TOTP MFA + Argon2id	Secure authentication
Validation	Zod	Schema validation on every endpoint

Layered View

L1 · Presentation

Web SPA Vendor Portal Research Portal Dashboards CAPI Mobile USSD/SMS Menus

L2 · API Gateway

Authentication Authorization (RBAC) Rate limiting Correlation IDs Audit intercept

L3 · Business Modules

M1 · Identity M2 · Projects M3 · Inception M4 · Results M5 · Finance M6 · Procurement

M8 · Beneficiary M18–M23 · Impact

L4 · Cross-Cutting Services

Identity & Audit Notifications Integration Administration

L5 · Data Layer

MySQL 8.4 Redis MinIO Longitudinal warehouse Anonymized store

04 Module Breakdown

PIIMS comprises 23 interconnected modules. Each module is a self-contained feature with its own data model, API surface, UI, and tests.

Core Business Modules

M1

Identity & Access

Users, roles, permissions, audit trail. Provides authentication, MFA, and hash-chained audit logs.

M2

Project Registry

Master register of projects and pipeline. Auto-generated project codes, stage gates, and pipeline tracking.

M3

Inception Reports

Report submission, versioning, review workflows, World Bank no-objection tracking.

M4

Results Framework

Components, sub-components, indicators, baselines, targets, and traffic-light status.

M5

Financial & Budget

Budget tracking, disbursements, FMIS sync, cost-per-output analysis.

M6

Procurement

Procurement plans, contracts, supplier performance, delay alerts.

M7

Supplier Portal

Isolated portal where suppliers submit deliveries, invoices, and warranties against their own contracts.

M8

Beneficiary Registry

Beneficiary registration with deduplication, consent capture, service delivery tracking.

M9

GIS / Geo-Spatial

Interactive maps, beneficiary and intervention markers, location verification.

M10

Document Management

Document upload, versioning, preview, access control.

M11

Reporting Engine

Standard and ad-hoc reports in PDF, Excel, CSV. World Bank ISR/PPMR export.

M12

Dashboards

Role-aware KPI dashboards with drill-down from portfolio to indicator.

M13

Grievance Redress

Intake via web + SMS, assignment, resolution, SLA tracking, analytics.

M14

Notifications

20 bilingual templates, in-app + email + SMS delivery, event-driven dispatch.

M15

Integrations

10 external system adapters with circuit breakers, retries, and idempotency.

M16

Administration

User management, master data, workflow config, system health.

M17

Mobile / Offline

PWA with offline data capture, sync queue, conflict resolution.

Impact Measurement Suite

M18

Impact Indicators

Impact domains, indicators, composite indices, wave assignment, theory of change.

M19

Survey Builder

Sections, 14 question types, skip logic, XLSForm import/export, publish & lock.

M20

Field Collection (CAPI)

Offline-first mobile data collection with GPS, photo, and paradata capture.

M21

Data Quality (QA)

Automated QA rules, flags, back-checks, collector scorecards, high-frequency checks.

M22

Warehouse & Research

Panel-linked warehouse, tiered anonymization (L0–L3), DUA-controlled research access.

M23

Impact Analytics

Wave comparison, difference-in-differences calculator, disaggregation analysis.

> **Impact measurement by design.** PIIMS is not just a registry — it captures impact data at scale, preserves it for 20+ years, and enables rigorous evaluation through anonymized research datasets.

05 Core Entity Model

The data model comprises 62 entities across 14 domains. Every entity has a UUID primary key, audit timestamps, and classification metadata.

Domains Overview

Domain	Entities	Key Tables
Identity & Access	4	Organization, User, Role, AuditLog
Project Core	5	Project, Component, Subcomponent, Milestone, Location
Results Framework	2	Indicator, IndicatorValue
Financial	2	Budget, Disbursement
Procurement	4	Procurement, Contract, Supplier, Product, Invoice
Beneficiary & Service	3	Beneficiary, ServiceDelivery, Grievance
Documents & Reports	3	Document, InceptionReport, Report
Impact Framework	4	ImpactDomain, ImpactIndicator, IndicatorWave, TheoryOfChange
Survey & Questionnaire	8	Survey, SurveyVersion, Section, Question, Option, SkipLogic, Wave, Assignment
Field Collection	7	Collector, Supervisor, Submission, SubmissionAnswer, Paradata, MediaAttachment, Consent
Data Quality	4	QaRule, QaFlag, BackCheck, CollectorScorecard
Longitudinal Warehouse	3	PanelMember, TreatmentGroup, WaveDataset
Research Access	6	AnonymizationMap, ResearchDataset, DataRequest, DUA, ResearchProject, Publication
Cross-Cutting	4	Notification, SyncJob, MasterData, SystemSetting

Key Relationships

> **Project hierarchy:** Project → Component → Subcomponent → Indicator → IndicatorValue. All financial, procurement, and beneficiary records hang off this hierarchy, ensuring results, money, and people are always linked.

> **Impact chain:** ImpactIndicator → Question → Assignment → Submission → SubmissionAnswer → QaFlag → PanelMember → WaveDataset → ResearchDataset. Every impact data point is traceable from design to published research.

> **Audit chain:** Every mutation is written to AuditLog with a hash-chained record (prevHash + recordHash), making the audit trail tamper-evident.

06 Integration Architecture

PIIMS integrates with 10 external systems, each with a dedicated adapter, circuit breaker, retry policy, and idempotency guard.

Code	System	Direction	Protocol	Key Data
FMIS	MoF IFMIS	Bidirectional	REST + HTTPS	Budget appropriations, releases, expenditures
DBE	DBE Loan Mgmt	Bidirectional	REST + HTTPS	On-lending agreements, disbursements, repayments
WB	World Bank	Outbound	SFTP + REST	ISR/PPMR supervision reports
SMS	Telecom SMS	Bidirectional	SMPP + REST	Beneficiary SMS, delivery receipts
USSD	Telecom USSD	Bidirectional	REST	Feature-phone menu sessions
MM	Mobile Money	Bidirectional	REST	Payment initiation, callbacks
GIS	GIS Platform	Bidirectional	REST + WMS	Basemaps, boundaries, spatial queries
IdP	Government IdP	Inbound	OIDC / SAML	Federated authentication
ODK	ODK Central	Bidirectional	REST + XLSForm	Survey forms, submissions
Safe Room	Research platform	Outbound	SFTP	Anonymized research datasets

Resilience Patterns

Circuit Breaker

Opens after 5 consecutive failures. Prevents cascading failures when an external system is down.

Exponential Backoff Retry

Automatic retry for transient errors (network, 5xx, 429) with exponential delay.

Idempotency Keys

Every push is tagged with an idempotency key so retries don't create duplicates.

Graceful Degradation

When an integration fails, PIIMS continues operating. Sync jobs queue for later processing.

Webhook Endpoints (Inbound)

Endpoint	Source	Purpose
`POST /channels/sms/inbound`	SMS Gateway	Beneficiary commands (REG, CONF, GRV, STAT, STOP)

Endpoint	Source	Purpose
`POST /channels/sms/delivery-receipt`	SMS Gateway	Delivery status callbacks
`POST /channels/ussd`	USSD Gateway	Menu navigation sessions
`POST /collect/odk-webhook`	ODK Central	Field submission notifications
`POST /payments/callback`	Mobile Money	Payment status updates

07 Security & Access Control

PIIMS implements security at every layer — authentication, authorization, transport, storage, audit, and compliance.

Authentication

Control	Implementation
Primary auth	JWT access tokens (15 min) + rotating refresh tokens (8 h)
Password hashing	Argon2id (19 MB memory, 2 iterations, parallelism 1)
MFA	TOTP (RFC 6238) — mandatory for privileged roles
Account lockout	5 failed attempts → 15-minute lockout
Session revocation	Refresh tokens stored in DB, revocable on logout
SSO	OIDC / SAML federation with government identity provider

Authorization — RBAC

PIIMS uses **Role-Based Access Control** with the following properties:

- **43 roles** across 14 stakeholder groups
- **Fine-grained permissions** — 60+ permissions like `project:read`, `budget:write`, `impact:design`
- **Wildcard matching** — `*` for all, `project:*` for any project permission
- **Organizational scoping** — MoF officers see only their org's data
- **Geographic scoping** — Regional officers see only their region
- **Field-level masking** — beneficiary PII masked for non-authorized roles

Access Control Matrix (Sample)

Role	Projects	Finance	Beneficiaries	Impact
System Administrator	Full	Full	Full	Full
MoWE PIU Officer	Full	Read	Read/Write	Read
MoF Budget Officer	Read	Full	—	—
DBE Loan Officer	Read	Read/Write	—	—
World Bank TTL	Read + Approve	Read	Read	Read
Impact Evaluation Specialist	Read	—	—	Full
Data Collector	—	—	Own subs	Field only

Role	Projects	Finance	Beneficiaries	Impact
Supplier Admin	Own only	Own only	—	—
Local Gov Officer	Local only	—	Local only	—
Beneficiary	—	—	Own only	—
Researcher	—	—	Anonymized	Anonymized
Auditor	Read	Read	Read	Read

Data Protection

Data Class	Control
Public	No restriction
Internal	Authentication required
Confidential	RBAC + audit logging
Restricted	RBAC + encryption + audit
Sensitive Personal (SPI)	RBAC + encryption + audit + field masking
Regulated	Immutable storage + legal hold
Research (Anonymized)	Data Use Agreement + access log

Compliance

- **ISO 27001** aligned information security management
- **World Bank** data protection and safeguards requirements
- **Ethiopian data sovereignty** — data hosted in-country
- **WCAG 2.1 AA** accessibility compliance

08 Impact Evaluation Suite

PIIMS captures development impact — not just implementation. It measures whether interventions actually changed beneficiaries' lives.

The Impact Lifecycle

Design indicators → Build surveys → Collect field data

QA & validate → Warehouse & anonymize → Analyze & publish

Impact Domains

Income & Employment

Household income, days employed, enterprise revenue.

Energy Access

Hours of electricity, cooking fuel, energy expenditure.

Water Access

Distance to source, water quality, collection time.

Health

Clinic visits, illness episodes, child nutrition.

Education

Enrollment, attendance, literacy, study hours.

Gender

Women's decision-making, time use, asset ownership.

Environment

Fuelwood use, CO_2 avoided.

Resilience

Coping strategies, shock exposure.

Satisfaction

Service quality, complaint resolution time.

Waves & Panel Data

Wave	Purpose	Timing	Key Data
W0 · Baseline	Pre-intervention measurement	Project start	Full instrument, full sample
W1 · Midline	Interim measurement	Mid-project	Same instrument + additions
W2 · Endline	Post-intervention measurement	Project end	Full instrument
W3 · Ex-post	Sustainability measurement	1–3 years after	Core instrument

Anonymization Tiers

Level	Description	Techniques	Audience
L0	Internal — full identifiers	None (RBAC)	IE, Data Manager
L1	Pseudonymized	Hashed IDs, panel ID	Internal research
L2	De-identified	Generalization, top-coding	External researchers
L3	Aggregated	Suppression, aggregation	Open data

Difference-in-Differences

PIIMS computes DiD effects for treatment vs control groups across waves, with standard errors and p-values. Researchers can export the underlying anonymized dataset for their own analysis.

> **Impact attribution.** The treatment vs control comparison, combined with baseline and endline measurement, allows rigorous attribution of observed change to the intervention — meeting World Bank impact evaluation standards.

09 Technology Stack

PIIMS uses a modern, widely-adopted technology stack chosen for reliability, maintainability, and long-term support.

Frontend

Component	Technology
Framework	React 18 + TypeScript 5 (strict mode)
Build tool	Vite 5 + vite-plugin-pwa
Routing	React Router v6
Server state	TanStack Query v5
Client state	Zustand
Styling	Tailwind CSS 3 + CSS custom properties
Forms	React Hook Form + Zod
Internationalization	react-i18next (English + Amharic)
Maps	Leaflet + react-leaflet
Icons	Lucide React
Testing	Vitest + Testing Library + Playwright

Backend

Component	Technology
Runtime	Node.js 20 LTS
Framework	Express 4 + TypeScript 5 (strict mode)
Database	MySQL 8.4 (utf8mb4, port 3307 in dev)
ORM	Prisma 5
Cache	Redis 7
Queue	BullMQ (Redis-backed)
Storage	S3-compatible (MinIO for dev)
Authentication	JWT (jsonwebtoken) + TOTP (otplib)
Password hashing	Argon2id
Validation	Zod
Logging	Pino (structured JSON)

Component	Technology
Reporting	PDFKit, ExcelJS
Email	Nodemailer
Testing	Vitest + Supertest

Infrastructure

Component	Technology
Containerization	Docker (multi-stage builds)
Orchestration	Docker Compose (dev/staging), Kubernetes (production)
Reverse proxy	nginx + Traefik
CI/CD	GitHub Actions
Monitoring	Prometheus + Grafana
Logging	Pino → Grafana Loki / ELK
Backup	Automated MySQL dumps + MinIO mirror

10 Deployment & Operations

Environment Topology

Environment	Purpose	Access
Development	Feature development, local testing	Development team
Test / QA	Automated and manual testing	QA team
UAT	User acceptance testing	MoWE, MoF, DBE, WB
Staging	Pre-production verification	Ops, QA
Production	Live system	Authorized users
DR	Disaster recovery failover	Ops only

Production Topology

Ingress

WAF / DDoS protection Load Balancer (nginx) TLS 1.3 termination

Application

3× stateless app nodes Autoscaling Health checks

Data

MySQL primary + replica Redis cluster MinIO cluster

Background Workers

Report worker Notification worker Sync worker

Operational Characteristics

Characteristic	Target
Uptime	≥ 99.5%
Recovery Point Objective (RPO)	≤ 24 hours (≤ 1 hour target)
Recovery Time Objective (RTO)	≤ 4 hours
Concurrent users	≥ 500

Characteristic	Target
API response (p95)	< 500 ms
Backup frequency	Daily at 01:00 EAT
Backup retention	30 days local, 90 days cold, 20 years for impact data

Monitoring & Alerting

- **Prometheus metrics** — request rate, latency, error rate, queue depth
- **Grafana dashboards** — system health, business KPIs, integration health
- **Alerts** — error rate > 5%, p95 latency > 2s, integration failures, queue backlog
- **Structured logging** — JSON logs with correlation IDs, PII redacted
- **Audit trail** — hash-chained, immutable, tamper-evident

11 Delivery Roadmap

PIIMS is delivered in four phases. Each phase ends with a working system, documentation, and knowledge transfer.

Phase 1	Phase 2	Phase 3	Phase 4
Foundation (Weeks 1–6)	Core Modules (Weeks 7–20)	Impact Suite (Weeks 21–32)	Polish & Handover (Weeks 33–38)

Phase 1 — Foundation

- Design tokens, component library, base app shell
- Authentication, RBAC infrastructure, role model
- Database schema (62 entities), migrations, seed data
- Express server, error handling, logging, monitoring
- CI/CD pipelines, Docker builds

Phase 2 — Core Modules

- **M2 Project Registry** — wizard, list, detail, pipeline
- **M3 Inception Reports** — submission, review, no-objection
- **M4 Results Framework** — components, indicators, traffic-light
- **M5 Financial** — budgets, disbursements, FMIS sync
- **M6 Procurement** — plans, contracts, delays
- **M7 Supplier Portal** — isolated supplier self-service
- **M8 Beneficiary Registry** — deduplication, consent, service delivery
- **M9 GIS** — maps, markers, geo-verification
- **M10 Documents** — upload, versioning, signed URLs
- **M11 Reporting** — PDF, Excel, CSV, World Bank export
- **M12 Dashboards** — role-aware KPIs
- **M13 Grievance** — intake, SLA, resolution

Phase 3 — Impact Suite

- **M18 Impact Indicators** — domains, composite indices, waves
- **M19 Survey Builder** — 14 question types, skip logic, XLSForm

- **M20 Field Collection** — CAPI, offline-first, GPS, paradata
- **M21 QA** — rules engine, flags, back-checks, scorecards
- **M22 Warehouse** — panel linking, anonymization, DUA
- **M23 Impact Analytics** — DiD calculator, wave comparison

Phase 4 — Polish & Handover

- M14 Notifications — 20 bilingual templates, in-app/email/SMS
- M15 Integrations — 10 external adapters, circuit breakers
- M16 Admin — user management, master data, audit
- M17 Mobile/Offline — PWA, sync queue
- End-to-end tests, performance tuning, security audit
- Complete documentation, runbooks, training, handover

Delivery Milestones

Milestone	Deliverable	Acceptance	Payment
M1	Foundation Demo	Login, RBAC, base shell	15%
M2	Core Modules Live	Project → Inception → Results flow	25%
M3	Financial & Procurement	Budget, disbursement, contracts	20%
M4	Impact Suite Live	Full indicator → warehouse → DiD	25%
M5	Final Handover	Training, docs, 90-day support	15%

12 Approval & Handover

Handover Package

Deliverable	Format	Purpose
Source code (frontend + backend)	Git repository	Full implementation, commit history
Database schema + seed scripts	Prisma migrations	Reproducible database
System Requirements Document	PDF	Complete requirements traceability
System Architecture & Design Document	PDF	Architecture decisions and design rationale
API Specification	OpenAPI + PDF	Interface contracts for integration
Data Management Plan & Data Dictionary	PDF	Governance and entity definitions
UI/UX Design Specification	PDF + Figma	Design system and prototypes
Deployment Runbook	Markdown	Step-by-step deployment and rollback
Operations & Maintenance Manual	PDF	Day-to-day operational procedures
Backup & Restore Procedures	Markdown + Scripts	Data protection and disaster recovery
Security & Compliance Documentation	PDF	Security architecture and audit trail
Training Materials per Role	PDF + Videos	43 roles, 13 role-specific manuals
Test Plan & Test Cases	PDF	QA verification and acceptance

Handover Activities

1. **Code walkthrough (2 days)** — architecture, module-by-module review with MoWE IT team.
2. **Deployment walkthrough (1 day)** — live deployment to staging together with ops team.
3. **Incident response drill (1 day)** — simulated failure and rollback exercise.
4. **Documentation review (1 day)** — walkthrough of RUNBOOK, SECURITY, CHANGELOG.
5. **Role-based training (5 days)** — separate sessions for each of 13 user groups.
6. **Transition support (90 days)** — HWO on-call during the initial production period, gradually reducing as MoWE IT builds confidence.

Support Commitment

Period	Support Level	Response Time
Week 1–2 post-handover	HWO daily on-site	≤ 2 hours (P1)
Week 3–4	HWO on-call every other day	≤ 4 hours (P1)
Week 5–8	HWO weekly check-in + issue triage	≤ 8 hours (P1)
Week 9–12	HWO bi-weekly	≤ 24 hours (P1)
Week 13+	Standard support SLA only	Per contract

Approval Sign-off

By signing below, the parties confirm that PIIMS has been delivered in accordance with the agreed specification, that all acceptance criteria have been met, and that responsibility for day-to-day operations is transferred to the client's IT team.

For HWO Software Development & ICT Consultancy

Project Director · Date: _____

For Ministry of Water and Energy

State Minister / CIO · Date: _____

For World Bank

Task Team Leader · Date: _____

For Ministry of Finance

Authorized Representative · Date: _____

> **System delivered.** PIIMS is a complete, production-ready, documented, and maintainable system — built to serve the Ministry of Water and Energy and the people of Ethiopia for the next decade.

H

HWO Software Development & ICT Consultancy

Enterprise solutions for governments and development partners

Services	Custom software development · System architecture · ICT consulting · Data engineering · Impact evaluation systems
Sectors	Government · Development finance · Energy · Water · Health · Education · Agriculture
Specializations	World Bank project systems · Bilingual platforms (English/Amharic) · Offline-first mobile · Longitudinal impact data
Delivery Method	Phased delivery · Full documentation · Knowledge transfer · Post-handover support
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